

NAME: _____

PERIOD: _____

DATE: _____

Two Analysts, One Planned Test

AP Statistics · Unit 3 · Topic 3.12 · B: 2026-12-15 · E: 2027-01-27

[STAR] TODAY'S PROBLEM

Suppose an online retailer has 50,000 active accounts. Before collecting data, its team asks, “Does a progress meter change the proportion of active accounts completing checkout?” It randomly selects 360 accounts and assigns 180 to each screen. The table shows the outcomes. Let p_M and p_S be the corresponding population completion proportions.

Nia: “Use $H_0 : p_M - p_S = 0$ versus $H_a : p_M - p_S \neq 0$ because either direction is a change.”

Omar: “Because $\hat{p}_M$ is larger, use $H_a : p_M - p_S > 0$.”

Checkout outcomes

Screen	n	Complete	Other
Meter	180	126	54
Standard	180	108	72

FIRST TAKE — before the video (5 min)

Which analyst’s setup should the team use? Cite one phrase or timing detail from the study.

[BOOK] SET THE QUESTION, PARAMETERS, AND CONDITIONS (keep on the page)

Define population proportions before writing hypotheses. For no difference, $H_0 : p_1 - p_2 = 0$. Choose $H_a : p_1 - p_2 > 0$, < 0 , or $\neq 0$ from the research question set **before** viewing results; do not choose a tail because of the observed sign. Use a two-sample z -test for a difference of proportions with independent random samples or randomized groups. Check each 10 percent condition when sampling without replacement. Under H_0 , pool: $\hat{p}_c = (x_1 + x_2)/(n_1 + n_2)$; all four pooled expected counts must be at least 10.

Finish after the video:

DOK 1 (a) Define p_M and p_S in context. Calculate $\hat{p}_M$, $\hat{p}_S$, and the observed difference $\hat{p}_M - \hat{p}_S$.

DOK 2 (b) Name the test procedure. Calculate the pooled proportion and verify randomization, the 10 percent condition, and all four pooled expected counts.

DOK 3 (c) Decide which setup is defensible and state the hypotheses. Use the word ‘change,’ the timing of the tail choice, condition evidence, and both kinds of randomization. Explain how the rejected setup would misrepresent the strength or scope of the evidence.

Frame: I would register H_0 : _____ and H_a : _____ because the pre-data question asks _____.

Frame: Changing the tail after seeing _____ would mislead readers by _____; the design supports _____.

EXIT — turn this in

One thing the video changed about my first take: _____